



PyCon JP 2026

How Python helped me getting the JLPT N1?

From dattebayo! to Leading an Engineering Team in Tokyo

Jose

SODA inc



**How many of you here today
speak Japanese?**

2016 (My only vocabulary)



俺 "ore"

Translation: "I" (Ultra-masculine)

⚠ USE WITH BOSS AT YOUR OWN RISK

だってばよ "dattebayo"

Translation: "Believe it!" (Naruto)

© NOT A REAL WORD, BELIEVE IT!

こんにちは "konnichiwa"

Translation: "Hello" (Normal word)

✅ SAFE TO USE IN SOCIETY

お兄ちゃん "onichan"

Translation: "Big brother" (Cringe at work)

💀 MAXIMUM SOCIAL AWKWARDNESS

2024 (The top)



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氏名 Name	JUANJOSE HAFID TENORIO PENIA
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レベル Level	N1
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**How I did it? And how did
Python helped me?**

Is English Really Enough?



"The 'official' language for IT engineers is English, right?"

Well, yes, but not quite like that here.

The Reality of English-Only Tech Roles



A Highly Concentrated Ecosystem

The market for pure English-speaking engineering roles in Japan is remarkably tight.

300–500

Total Companies

Total estimated firms in Japan accommodating English-only tech talent.

150–250

Active Startups

Startups actively hiring, heavily driven by Deep Tech and AI sectors.

Severe Logistical Friction

Despite an eagerness to onboard international talent, relocating engineers from overseas presents immense bureaucratic risks and high upfront costs.

40%

Of Job Postings

require that an applicant might already be in Japan

Not Convinced?



9.1 Work Visas

New JLPT N2 Requirements

Effective **April 15, 2026**, first-time overseas applicants must prove proficiency equivalent to **JLPT N2 (or CEFR B2)**.

This rule targets applicants at **Category 3 and 4 companies** (SMEs and startups) to prevent visa abuse..

9.2 Permanent Residency

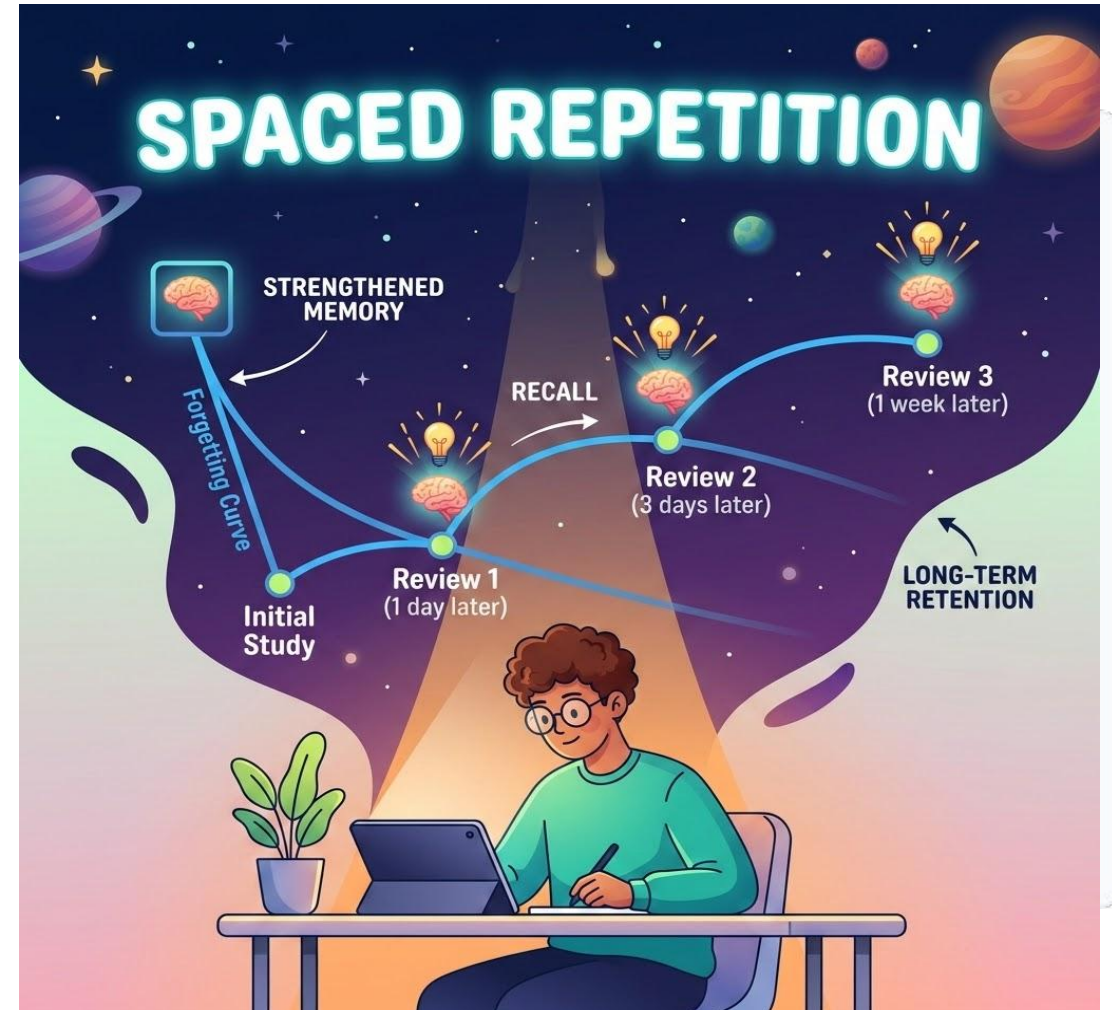
The Push for Mandatory Japanese

A formal **language requirement for PR** is being introduced via a mandatory **"Japanese Language and Lifestyle Learning Programme"**.

With **JLPT N2** as the discussed benchmark, the program will test skills and integration. Guidelines are expected by **FY2027**.

How Do We Learn?

- 🧠 **The Core Mechanic:** Psychologists have proven that reviewing information at dynamically expanding intervals halts the forgetting curve.
- 📅 **The Algorithm:** There's been a decade-long algorithmic competition to find the right interval: SM2 (1987), Free Spaced Repetition Scheduler (2024).
- ∞ **Permafrost Memory:** Some studies show recall spanning 50 years. Human memory forms deep, unshakeable roots under the right conditions, for some reason (?).





**Of course, we have the
most popular app and
influential program for this**







The beginning

JLPT N5 - N4 (2016-2018)

The Beginning: JLPT N5 & N4 (2016–2017)

JLPT N5: The Foundation

2016 • 4 Months Race

The easiest level is mostly focused on **symbol recognition**, including hiragana and katakana.

100 kanjis + grammar + listening

JLPT N4: Grammar

Jan 2017 – June 2017

Transitioning from isolated symbols to **contextual grammar understanding**.

Years of passive anime exposure transitioned from simple noise to a structured database of recognizable concepts.



JLPT N3: The Engineering problem

June 2017 – December 2017 The Context Deficit Challenge

At the time, I chased N3 believing it was a stepping stone to employment. However, the leap in volume was immense—equivalent to **N5 and N4 combined** (500 characters & 1000 words).

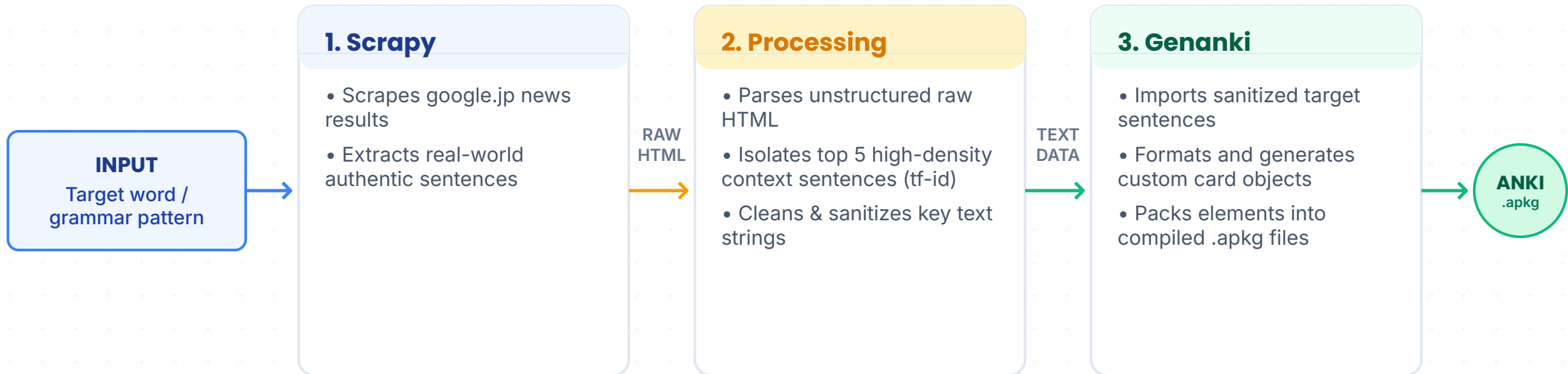
Separated from Japan, I lacked the natural environment to apply this vocabulary. **How do you scale context when you have no one to speak to?**



Enter Python



JLPT N3: Adding context





And I stopped

2018 – 2019



2018 – The Gap Year & Pivot

I spent the year working and preparing a high-stakes MEXT scholarship application.

While the MEXT scholarship application ultimately failed, my previous tutor connected me directly with a Japanese CEO.

I secured an interview—this was the defining moment when intermediate conversational N3 fluency paid off, allowing me to speak and make myself understood.



Japan

My first job (Sep 2019)



Not good enough

N3 vocabulary was good enough for communication, but not for a fully Japanese environment.

I figured I had to get the N2 to get even more words to my brain.

I lacked technical words (*yogo*) required for writing specifications and discussing with product managers.

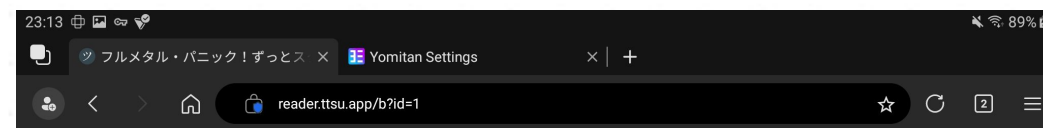


JLPT N2 (Jan 2020 – Dec 2020)

Immersion for N2 (Jan 2020–Dec 2020)

N2 brings **long lectures, specialized grammar**, 1500 words and 1000 Kanjis.

- My old pipeline was working fine for kanjis and grammar but still it was hard to maintain. so I tried the community supported add on **Yomichan**. It would generate Anki flashcards for any content I would see on the web.
- Being immersed tremendously helped especially on the reading and listening sections.
- In the professional aspect, I would learn a lot of **technical stuff**..



いく
んでい
して、
全国
朝か
らない。
め、風
あの
英語で

てきた。世界史では、出るわけないとタカをくくって
いた古代南米史がどっさり出てしまった。そして古文
では、よりにもよって源氏物語が出た。

特に源氏物語！ あんな、一〇〇〇年前の恋愛至上
主義の女が書いた妄想——ヤリチン男の腐りきった女
遍歴なんぞが理解できるかどうかなかなかで、現代に生
きる若者の人生を決めないで欲しい！ ……と信二は
つくづく思う。

英語
CNC100 526 VNFreq 3521 JPDB 4620, 104458
Anime & J-drama 2662 BCCWJ-LUW 912
アクセント辞典 えいご [0] ●▼▼
デジタル大辞泉 えい - ご【英語】
インド - ヨーロッパ語族のゲルマン語派に属する言語。現
在、大ブリテンと北アイルランドのほか、アメリカ・カナ
ダ、インド・オーストラリア・香港（ホンコン）などで使わ
れ、国際的な傾向が強く残っている。語彙の面で、フラン
ス語からの影響が大きく、ドイツ語・オランダ語などと親近
関係にある。この語族の言語の中では、最も孤立語的な傾
向を示す。
三省堂国語辞典 第八版 **えいご【英語】《名》
①イギリス・アメリカをはじめとした地域で話される、勢
力の大い言語。
②「英語①」を学ぶ教科。英語科。
朗読学習用国語辞典 第五版 えいご【英語】
文字が並
あつたり
容は変わ
切れを眺

プロローグ



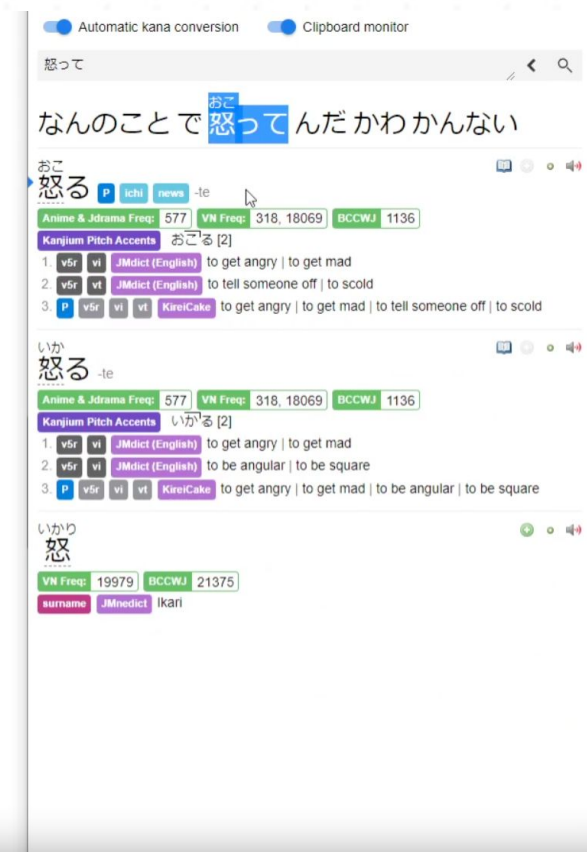
And I stopped

2021 – 2023 (The community moves forward)

Manga-OCR & Transformers

The Python community introduced [manga-ocr](#). Built on a Vision Encoder-Decoder transformer framework, it accurately parses vertical text, furigana, and stylized fonts from images, seamlessly linking native reading to flashcard creation.

The dataset was provided from a compilation of free mangas by the University of Tokyo (manga-109 dataset)



And chappy arrived



ChatGPT

Conquering JLPT N1 (Jan 2024 – July 2024)

Vocabulary & Kanji

Anki Was the Boss

Systematically brute-forced and memorized all **2,136 Joyo Kanji**. Constant spaced repetition formed the core structural foundation of the vocabulary stack.

Grammar Synthesis

Chat GPT Integration

Used as the primary sentence generator. Since the Model Context Protocol (MCP) was only in its early conception, the pipeline relied on direct trust in Chat GPT's raw linguistic outputs.

I would run a script with genanki to generate the flashcards.

Comprehension

Organic Reading & Listening

Allowed comprehension to develop naturally. Trusted the permafrost and used physical books as baseline indicators to measure visual reading speed and pacing over time.

December 2024 (The end)

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So it was over...

A close-up of a green cartoon owl's face. The owl has large, white, almond-shaped eyes with black pupils. Its beak is orange and slightly curved. The owl's face is a vibrant green color with some lighter green shading around the eyes. The background is dark green.

Not really

The illusion of mastery

The Recall Deficit

I possessed great recognition for kanji characters when reading, but froze when forced to produce them manually.

The Pronunciation Blindspot

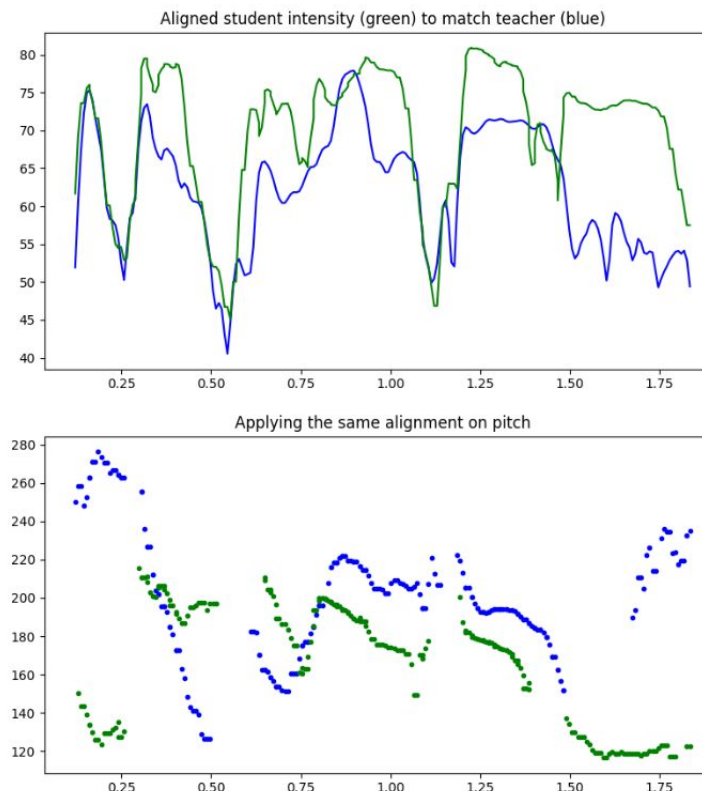
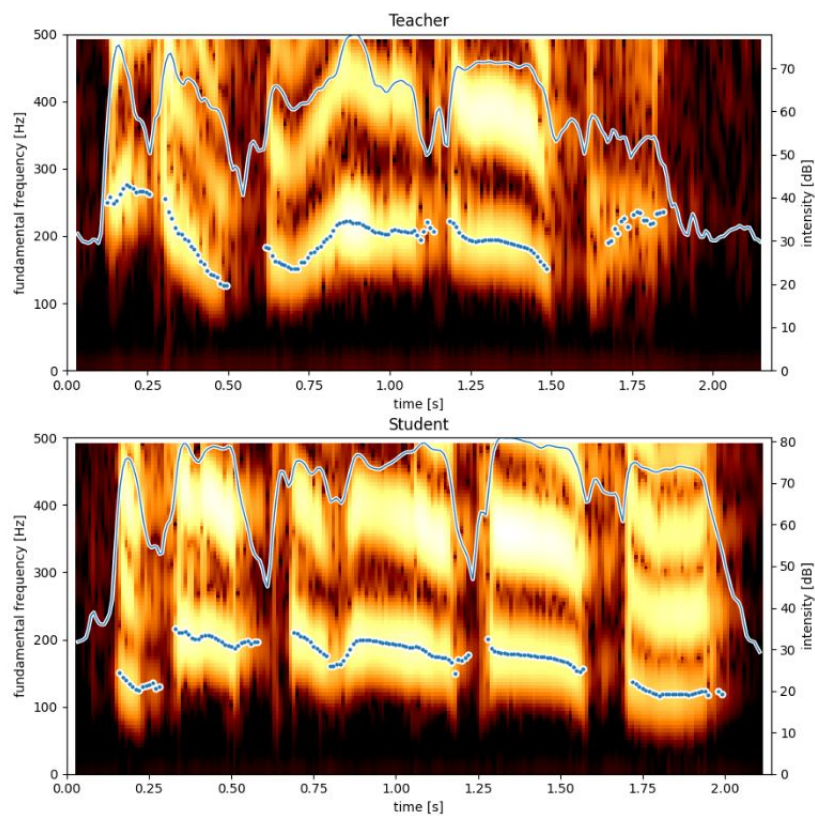
Despite continuous speaking, I remained entirely oblivious to the fundamental importance of pitch accents.

My partner would constantly point out at my accent mistakes.

**Do you need a
Japanese waifu to
pronounce better?**

Onsei

<https://github.com/itsupera/onsei>



The Pronunciation gap



Not really

Onsei

An ongoing project. It records your voice, analyzes the pitch contour, and grades your cadence against the standard Tokyo dialect.

NHK Pronunciation Dictionary

The gold standard. While less automated, manually verifying pitch accents against the official NHK database provides 100% reliable native audio.

The 2025/2026 Learning Stack



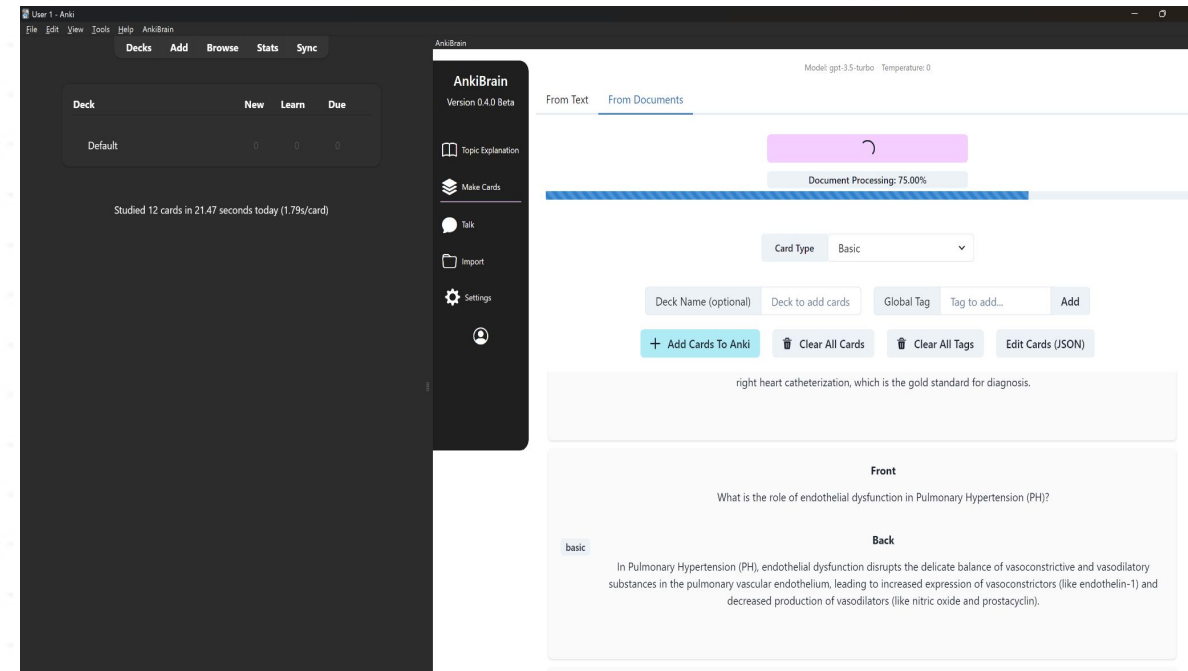
Claude Code & Genanki: Instead of simple scrapers, use agentic LLMs to parse specific datasets (like industry buzzwords) and automatically construct `.apkg` files.



Anki Brain: Advanced integrations now allow you to feed entire PDFs, technical ePubs, or corporate documents into an AI, extracting the vocabulary instantly.



Organic Returns: Eventually, the goal is to step away from the flashcards and let the language become second nature through pure consumption.



A never-ending journey



- ✓ **Organic is King:** Aim for organic learning. It's more fun that way.
- 👂 **Recognition vs Recall. :** Living in the country makes listening second nature but it's trickier to learn pronunciation. You can recognize kanjis but cannot write them (even the Japanese cannot do it sometimes).
- 📈 **The Professional Multiplier:** Having reliable, business-level Japanese is a massive career booster: broad interview access, negotiation edge, and more friends :)



**Will this matter in
50 years?**

I believe so. Language is your ultimate insurance.

Questions?

Thank you for listening!

<https://github.com/jota12x/pycon-jp-2026-docs>



[We are hiring!](#)